

Course Report

(2025)

Academic Year

....2024....
/...2025.....

Semester

Spring

1. Basic Information

Course Title (according to the bylaw):	Probability and Statistics
Course Code (according to the bylaw):	BES 299
Department/s that participated in the teaching:	Basic Science Engineering
Total number of credit hours/points of the course:	3
Course Type:	Core
The level to which the course was introduced:	Level 2
Academic Program:	Credit programs
Faculty/Institute:	Faculty of Engineering
University/Academy:	Menoufia University
Name of Course Coordinator:	Prof. Dr. Ahmed Al Sawy
Course Report Approval (Attach the decision/minutes of the department /committee/council)	
Date of approval of the course report	6/1/2025

2. Data and Statistics

Course Instructors			
Number of Faculty Staff		Number of Teaching Assistants	
Full-time (at least 4 working days)	Part-time (1 or 2 days)	Full-time (at least 4 working days)	Part-time (1 or 2 days)
4	1	4	1
Instructor Name	Department	Academic degree	Specialty
1. Ahmed Al Sawy	Basic Science engineering	Professor	Mathematics
2.			
Notes (if any):			

Teaching and Learning					
Number of weeks of the actual study	Total number of Theoretical teaching hours (Lectures/)	Total number of training hours (practical/clinical/...)	Total number of field training hours (if any)	Total number of self-learning hours (if any)	Other (to be mentioned)
13	13	-	-	-	-
Notes (if any) on: Topics not covered, changes in teaching methods, number of teaching hours or content:					

Student Assessment Methods that have been Implemented				
Method of assessment *	Date of Evaluation	Marks/ Score	Type and number of questions	Measured Course Learning Outcomes (Mention the text)
Exam (1) Semester work	2/25/2025	5	Problem Solving, 1	a1-1) Identify types of statistics and sampling. a1-2) Define the axioms of probability, statistical independence, and the pay theorem. a1-3) Develop a continuous distribution function.
Exam (2) Semester work	5/2/2025	10	Problem solving m, 2	a10-1) Define point estimation and the central limit theorem, as well as interval estimation.
Midterm exam	3/26/2025	20	Problem solving m, 2	a2-1) Develop a continuous distribution function . a2-2) Develop discrete probability distributions (uniform, binomial, geometric, Poisson). a2-3) Use statistical analyses and objective engineering for Continuous probability distributions (normal, exponential, gamma, and Weibull)
Final Practical Exam	Click or tap to enter a date.			
Final Oral Exam	Click or tap to enter a date.			
Final written exam	5/25/2025	50	Problem solving m, 4	a1-1) Identify types of statistics and sampling. a1-2) Define the axioms of probability, statistical independence, and the pay theorem. a10-1) Define point estimation and the central limit theorem, as well as interval estimation. a10-2) Apply statistical software for statistical analysis.

				a10-3) Define and apply the Hypothesis Tests.
Field training	Click or tap to enter a date.			
Projects/ Assignments/ Portfolio/ Logbook/ ...	5/13/2025	15	Problem solving m, 2	All attitudes.
Other (to be mentioned)	Click or tap to enter a date.			
* The methods mentioned are examples the institution may add and delete depending on the nature of the course Comment on the procedures and rules of student assessment that have been implemented (e.g. examination committees, exam paper review, blueprints,), and the rules for passing the course:				

Student Assessment Results								
Number of students (who started the course):				47				
Number of students (who completed the course/ sat for the exam):				47				
Number of students who did not attend the final written exam (absent - deprived - withdrawn):				47				
Total number of students who passed the exams successfully:				43				
Percentage of success (out of the total number of students who sat for the final exam)				91%				
Grade Distribution *								
Grade	A+	A	B+	B	C+	C	D+	D
Number of students	2	14	7	31	1	1	1	5
Percentage	4	30	15	27	2	2	2	11
* In case the distribution of grades differs from the current model, it should be replaced by grades according to the bylaws								
Total number of students who failed the exams:				4				
Percentage of failure (for the total number of students who took the final exam)				9%				

Commenting on student results, analyzing student performance (including factors that affected the results, if any), and comparing them with the results of the previous year:

- **Course LOs:** Shows the percentage of students achieving a “Good” level on various Learning Outcomes (a1-1 through a10-3). The A1 group has a higher rating.
- **Program LOS:** Overall “Good” rating for Level A Program Learning Outcomes, with a 79.4% achievement.
- **Grade Distribution:** Shows the number and percentage of students receiving each letter grade.

Strengths:

- **Good Achievement of Learning Outcomes:** Most students (between 78.4% and 80.3%) achieve a “Good” level on the specific Course Learning Outcomes.
- **Level A program is followed:** The Level A program is followed, at a good rating, which is also a high level.

Weaknesses:

- **Lack of Granularity in LOS Data:** Like the previous analysis, we lack detail about the “Good” rating. We don’t know the distribution of scores within that range, the assessment methods used, or the criteria for “Good.” If they all score 70%, that’s a low good score.
- **Not a lot of grade variation:** There are a lot of similar grades at different ranges, which is fine, but indicates not many high grades.
- **At the low end:** These grades should be considered, as they all account for 9%. Since it is the last phase, it means that these students did not attend after the exam.

Specific Recommendations:

1. **Gain More Qualitative Feedback:** Implement short, anonymous surveys to students to provide qualitative feedback for students with bad grades.
2. **Gain Access to the assessments and grades, to review the actual assessments:** Since there are no more details, you cannot see a more accurate result. If there are tests that are not fair, grades will be similar.
3. **Re-evaluate with a different standard:** Try to rate those grades with a curve rating, to see if you get a better result.

Course Quality Evaluation - Students' Evaluation of the Course:

Analysis of Survey Version 1 (3 Respondents):

- **Tiny Sample Size:** The most significant weakness is that only 3 out of 3 students completed the survey. This makes the data highly unreliable and not representative of the class.
- **Overall Rating: 3.4:** This suggests a generally positive view, but given the small sample, it could be skewed by the opinions of just a few individuals.
- **Limited Variation:** The ratings are mostly clustered around 3.7, with no strong areas of either strength or weakness.
- **Lack of Comments:** The absence of student comments further limits our understanding.

Strengths (Version 1):

- All Students Responded.
- The data looks consistent (although clustered).

Weaknesses (Version 1):

- The most significant weakness is that only 3 students are rating.
- The ratings provided don't show any specific areas for improvement.

Analysis of Survey Version 2 (38 Respondents):

- **Improved Sample Size:** 38 out of 45 students completed the survey, resulting in an 84.4% response rate, which is much better.
- **Overall Rating: 4.3:** indicates that the students generally liked the subject.

Strengths (Version 2):

- Data shows a very high number of ratings.

Weaknesses (Version 2):

- Lacks more specific grades other than very positive.

Comparative analysis:

Compared to the old list:

- There's now the same number of high and low ratings.

- This test shows generally more consistent results.

Conclusion

You must only use the grades from the survey with more ratings. It is essential to improve the ratings to achieve better and more equitable results.

3. Student Feedback *

* Feedback from students must include their evaluation of the following: scientific content – teaching and learning methods – facilities and learning resources – examinations -
(attach the questionnaire analysis report or any other means used, and the points evaluated).

Item	Comment
Means of Evaluation:	تقرير الاستبيان
Timing of Evaluation:	شهر
Number of students who participated in the course evaluation	12
Percentage of participants to the total number	%70
Important points of satisfaction	1. Clarity and Understandability: Students are satisfied when the course material is presented clearly and is generally understandable. 2. Relevance to Specialization: A strong point of satisfaction is when students perceive the course content as relevant to their field of study or future careers. 3. Inclusion of Modern Information and Practical Examples: Students appreciate courses that incorporate up-to-date information and provide practical examples that illustrate the application of concepts. 4. Meeting Expectations: When the course meets the general expectations students have for it, this contributes to satisfaction. 5. Achieving Learning Outcomes: Students are satisfied when they feel the course successfully helps them learn and achieve the intended

Item	Comment
	learning outcomes, especially when they achieve a "Good" or higher rating. 6. Availability of Study Resources: Students are satisfied when instructors provide useful study resources (like practice exercises) that aid their learning. 7. Course Excitement/Engagement: While often a weakness, when a course <i>is</i> perceived as exciting or engaging, it's a significant point of satisfaction, although this was noted as low in some analyses.
Important points of dissatisfaction	• Unclear Explanations: When the instructor struggles to explain concepts clearly or uses confusing language, students become frustrated. • Unresponsiveness to Questions: A lack of willingness or ability to answer student questions thoroughly. • Lack of Enthusiasm or Engagement: Students are dissatisfied when the instructor appears unenthusiastic or disconnected from the material. • Inaccessibility: If the instructor is difficult to reach outside of class hours or unresponsive to emails, this creates dissatisfaction. • Unfair Assessment: Inconsistent or unfair grading practices are a significant source of dissatisfaction. • Lack of Feedback: Insufficient or unhelpful feedback on assignments or exams can hinder student learning and decrease satisfaction. • Poor Communication Skills: Students are dissatisfied when instructors are difficult to understand, unorganized, or present information in a confusing manner.

4. Instructors' Reflection *

* Instructors' views on the educational process, scientific content, adequacy of resources, etc. (if any)

3. Course Enhancement

Comment on uncompleted corrective/improvement actions from last year's plan (if any, mentioning the reasons)

Course development plan for the next academic semester/year

(considering the student assessment results as well as the student feedback and instructors' reflection)

Action Plan for next Semester / Academic Year AD

Goals:

- Increase student engagement and enthusiasm for the course material.
- Enhance specific areas of the course that students perceive as less strong.
- Maintain the course's strengths and build on them.
- Increase survey response rates in the future.

No.	Points that need development or improvement	Corrective/ Improvement Actions	Methods of implementation	Notes
1.	Investigation and Assessment	Review Course Materials	Examine the syllabus, lecture notes, assignments, and assessments to identify areas that may be outdated or less engaging.	
2.	Resources Needed	Time for course review and revision	Access to learning management system (LMS) for posting announcements and survey	
3.	Measurement and Evaluation	Increased Student Engagement	track the grades of students who have bad	

			results, and if they improve.	
4.				

**Name and Signature
Course Coordinator**

Prof. Dr. Ahmed Al Sawy

**Name and Signature
Head of the department council**

Prof. Mohamed Eltabey Mohamed Elsayed